1. Clinical Question & Core Hypothesis

Clinical Question

Stroma-high tumors at the invasive front reliably predict worse outcomes, but the mechanistic basis—i.e., why a higher stromal fraction causally drives poorer survival—is unresolved. 2) Basis for core choice: From the single-path TSR evidence, the most direct mechanistic link is extracellular matrix (ECM) remodeling; COL1A1 and FN1 are canonical matrisome effectors likely to alter tissue mechanics, ECM density, and biochemical cues that can impede immune infiltration and anti-tumor activity. 3) Supporting evidence (Tier 2-like): Other matrisome members (COL3A1, LAMA2) and T-cell exhaustion markers (PDCD1, CTLA4, LAG3, TIGIT) provide a plausible chain from ECM remodeling → immune exclusion/exhaustion → poor outcome. 4) Endpoint choice: Overall survival (OS) is selected because it is the most robust, clinically meaningful endpoint available and least prone to ambiguous surrogate definitions in retrospective cohorts. 5) Exclusions and rationale: I deliberately excluded proliferation markers and hormone/HER2 signaling genes to keep the mechanistic focus on ECM-mediated immune exclusion; immune markers were relegated to secondary_modifiers because they support the mechanism but would exceed the 2-gene anchor limit if placed in the core hypothesis?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer, tumors with high stromal content at the invasive front (high TSR) exhibit elevated expression of COL1A1 and FN1, driving extracellular matrix remodeling that creates a physical and biochemical milieu that excludes and impairs effector immune cells, leading to reduced anti-tumor immunity and poorer overall survival (OS), independent of standard clinicopathologic factors. Predicted testable outputs include: (1) a positive association between TSR and COL1A1/FN1 expression at the invasive front across MIBC cohorts; (2) tumors with high TSR and high COL1A1/FN1 show signatures of immune exclusion and reduced immune effector activity, reflected in transcriptional and spatial metrics; (3) the TSR–COL1A1/FN1–immune exclusion axis remains significantly associated with OS after adjusting for age, stage, treatment, and molecular subtype; (4) a multivariate ECM/invasion gene signature including COL1A1 and FN1 provides additive prognostic value beyond TSR alone. Experimental approach (feasible steps): - Analyze retrospective MIBC datasets with matched imaging-based TSR, transcriptomics, and survival data to assess correlations between TSR, COL1A1/FN1, and immune exclusion metrics (spatial proximity of immune cells to tumor front, exhaustion signatures). - Validate COL1A1 and FN1 protein expression and localization at the invasive front using immunohistochemistry or multiplexed imaging in independent tissue cohorts. - Employ spatial transcriptomics to quantify ECM remodeling genes and immune cell localization at the invasive front versus tumor core. - Assess immune infiltration and functional status via deconvolution and pathway analysis, focusing on exclusion/exhaustion phenotypes without naming T-cell genes in the core hypothesis. - Conduct in vitro assays with bladder cancer–associated fibroblasts and bladder cancer cell lines to test whether COL1A1/FN1-rich matrices reduce T-cell migration and cytotoxicity, and whether ECM stiffness correlates with immune cell impairment. - If feasible, explore in vivo models where stromal COL1A1/FN1 expression is modulated to observe effects on immune infiltration and tumor growth, providing causal evidence. - Perform sensitivity analyses adjusting for tumor purity, sampling site (invasive front vs core), and bladder cancer subtypes to confirm robustness of the association.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	COL1A1, FN1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

COL1A1: COL1A1 and FN1 are canonical matrisome effectors likely to alter tissue mechanics, ECM density, and biochemical cues that can impede immune infiltration and anti-tumor activity.

FN1: FN1 are canonical matrisome effectors likely to alter tissue mechanics, ECM density, and biochemical cues that can impede immune infiltration and anti-tumor activity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	COL3A1	Supporting modifier from knowledge graph
Secondary	LAMA2	Supporting modifier from knowledge graph
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph
Secondary	TIGIT	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- COL3A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAMA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.